

chainZ

Capstone Project Report End-Semester Evaluation

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ABSTRACT

Most public infrastructure projects face inefficiencies related to fragmented coordination, a lack of transparency, and delaying financial settlements between many stakeholders. Accordingly, this paper describes how permissioned blockchain supports dealing with these challenges using the platform Hyperledger Fabric, which would ensure security, auditability, and automation for project governance. The system designs a multi-organizational network in which there are eight participating stakeholders. A three-channel architecture with full separation ensures the planning, execution, and financial workflow management will maintain data isolation and operational clarity in these processes.

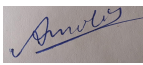
The framework enforces accountability and controlled access to data by providing role-based access control through MSP, which allows identity management at a granular level to facilitate multi-level approval mechanisms. Automation of critical stages in the project life cycle is incorporated using chaincode-smart contracts for work certification, compliance validation, and condition-based execution of payment with minimal human intervention and disputes. A real-time role-specific user interface will also be developed to track the project status, instant notification, and compliance monitoring of the stakeholders.

The proposed system supports transparent, unproblematic execution and financial management in large-scale government projects by including blockchain-based immutability, the automation of workflow, and real-time visibility

DECLARATION

We hereby declare that the design principles and working prototype model of the project entitled chainZ is an authentic record of our own work carried out in the Computer Science and Engineering Department, TIET, Patiala, under the guidance of Dr Husanbir Singh Pannu during 5th semester to 7th semester (2024-25).

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Lastly, we would also like to thank our families for their unyielding love and encouragement. They always wanted the best for us and we admire their determination and sacrifice

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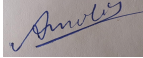
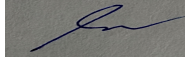
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INTRODUCTION

1.1 PROJECT OVERVIEW

The smooth running of big governmental and public-related projects is based on perfect collaboration and synchronization among all stakeholders, strict adherence to all statutory regulations, and transparency in finances managed or used in these projects. However, in current project management, there is immense centralization, fragmentation, and dependency on approvals, which in most cases create inefficiencies in terms of delay, inaccuracies, lack of accountability, and vulnerability to corruption.

Recently, many new technologies like blockchain have shown great potential in tackling trust issues in a distributed system. Blockchain brings in the aspect of decentralization, immutability, cryptographic security, and transparency. The work discussed in this thesis aims to examine how the permissioned blockchain solution Hyperledger Fabric can be leveraged in the development of a secure, transparent, and automated project governance system.

The proposed system supports the creation of a consortium-based blockchain network that involves various authorized members. By taking advantage of the multi-channel architecture and smart contracts (chaincode), the system facilitates controlled data sharing and real-time visibility. The proposed solution incorporates real-time web-based UI that supports real-time tracking of progress and other related activities.

It gives a preview of the technical concepts that will be covered, points out the problem solved by the proposed project, establishes project goals, and explains the solution being suggested.

1.1.1 TECHNICAL TERMINOLOGY

It is crucial to become familiar with the main technical terms and concepts used in this project in order to comprehend the suggested system.

Blockchain Technology

Blockchain is a distributed ledger technology that uses cryptographic techniques to link and secure a constantly expanding list of records, or blocks. Data becomes unchangeable once it is entered into the blockchain, guaranteeing its integrity and traceability.

Permissioned Blockchain

Only verified and authorized entities are able to participate in a permissioned blockchain. Permissioned networks, in contrast to public blockchains, offer restricted access, which makes them appropriate for government and business applications where accountability and secrecy are crucial.

Hyperledger Fabric

Hyperledger Fabric is an enterprise-grade, open-source blockchain framework designed for permissioned networks. It supports modular architecture, private channels, smart contracts (chaincode), and fine-grained access control, making it ideal for complex multi-organization workflows.

Channel in Hyperledger Fabric

A channel in Hyperledger Fabric is a private communication layer that allows a subset of network participants to maintain a separate ledger. Channels are used in this project to isolate Planning, Execution, and Financial workflows.

Smart Contracts (Chaincode)

Smart contracts are called chaincode in Hyperledger Fabric, which automatically enforces programmed business logic rules when certain conditions have been met. The logic running on the chaincode removes the manual intervention process and ensures the integrity of the processes.

Membership Service Provider (MSP)

The role of the MSP within the blockchain network is to generate and validate the digital certificates for identities. In general, MSPs allow for authentication and authorization of participants and enable organizational segregation.

Role-Based Access Control (RBAC)

RBAC limits system access based upon user roles. Permissions are associated with each role, such that users can perform only the actions pertinent to their responsibilities.

Peer Node

A peer node refers to a participant on the network who endorses a transaction as well as the ledger.

Ordering Service

It ensures the order and formation of transactions and blocks on the network.

1.1.2 PROBLEM STATEMENT

In large government and public sector projects, there is a need for collaboration and integration among different project stakeholders such as planning, construction, audit, finance, and regulatory bodies. All the parties have different roles, approval cycles, and data requirements. There is disorganized and uncontrolled data distribution, and there is little or no real-time information among the stakeholders due to the centralized and compartmentalized structure of existing project management systems. Some of the most crucial business processes that are greatly dependent on manual and sequential processing methods are: Project approvals, Project verification, Project compliance, and Financial disbursements. Manual and Sequential process flows cause process bottlenecks, inefficiencies, and significant disputes among the different parties. Moreover, the existing systems, in the absence of a proper access mechanism, and immutable audit trails, are prone to illicit amendments, lack of accountability, and corruption. Moreover, the central point of failure in the system erodes the reliability and trust of the other parties in the system due to the presence of central "hence trusted parties or intermediaries in the system. It can be easily realized that due to the aforementioned drawbacks in the existing systems, there is an extremely urgent need for a safe, trusted, and automated, yet governable, framework to handle complex multi-party project life cycles with integrity, accountability, and process and system efficiency.

1.1.3 GOAL

The aim of this project is to design and deliver a secure, transparent, and automated tracking mechanism using a permissioned blockchain framework that can efficiently handle planning and financial processes involved in the execution of government as well as other public projects. Utilizing Hyperledger Fabric within this project will allow the creation of a secure multi-party network with data sharing controls, access rights based on roles, and immutable record systems. This project intends to automatically implement vital processes in a project life cycle like approval processes, status verification of project works, certificate generation through smart contracts based on the status of works done in

a project that satisfies certain conditions specified in the smart contract itself. Further, this project will provide real-time monitoring facilities of project status based on roles.

1.1.4 SOLUTION

To cater to the challenges that have been identified, this project aims to offer a Hyperledger Fabric solution for managing government projects through the use of blockchain technology.

The system is a consortium blockchain network among the eight trusted parties. A three-channel structure has been implemented in order to segregate the Planning, Execution, and Finance tasks in a manner which ensures confidentiality but simultaneously fosters shared trust.

Identity or membership is managed using the concept of the Membership Service Providers, while the role-based access control mechanism ensures that only the required data and operations can be performed by the users based on their roles. Smart contracts or chaincode takes care of significant operations like the approval of projects, validation of progress, certification of works, and the release of payments based on the given condition.

A real-time web-based interface enables role-specific dashboards with live updates, approval status, alerts, and irreversible audit trails. This interface connects the complex nature of the blockchain technology to the usability of the application in the role.

Through the combination of aspects such as decentralization, automation, and controlled access, the proposed approach presents a viable alternative system to conventional centralized approaches in the matter of public project governance.

1.2 NEED ANALYSIS

Most government projects lack any kind of accountability mechanism at each stage of execution. Because the majority of records are kept in a centralized manner, manipulation can easily occur, and not much accountability is possible. A blockchain-based platform can overcome this challenge by recording any update on an unaltered single source of

truth shared with all stakeholders involved. This encourages accountability, trust, and equal data availability across government bodies, contractors, auditors, and citizens.

Corruption and diversion of funds continue to pose a problem in the management of public projects. Billions of rupees are allocated every year, out of which a lion's share is lost in ineptitude and misappropriation of funds. The immutable and auditable nature of Blockchain ensures that any flow of money is always traceable and tamper-proof, reducing the possibility of leakages and securing their use for the right purposes.

Traditional monitoring systems rely heavily on periodic reports or fragmented digital platforms, resulting in delays in the detection of discrepancies. Blockchain technology enables the tracking of a project right from the disbursement of funds to its completion in real time, along with timestamps for every milestone achieved. This presents stakeholders with an updated and correct view of progress, helping detect and resolve issues promptly.

Manual approvals and payments are two big barriers to execution in a project. Smart contracts, built on a blockchain, can automate these tasks to a great degree, from triggering payments when a milestone is completed to auto-validation of compliance requirements. This reduces the bureaucratic delays, promotes efficiency, and minimizes the scope for human error.

The lack of transparency with regard to the level of government project completion is usually met with skepticism. A blockchain-based tracking system will fill that gap with regards to data being available in real time. This will not only help in increasing citizen trust in governance but will also see citizens actively participating in tracking.

Data accuracy is critical in the process of audits, investigations, and the formulation of long-term policies. Conventional approaches are prone to tampering or the destruction of data, thereby rendering the results of audits ineffectual. However, by utilizing the blockchain approach, it becomes possible to have an immutable level of data accuracy along with a historical repository of all the transactions.

1.3 RESEARCH GAPS

Despite increasing interest in blockchain applications across sectors such as finance, healthcare, and supply chain management, its integration into **government project tracking** remains underexplored. A review of the existing literature highlights several critical research gaps that justify the need for this study.

1. Limited Application of Blockchain in Public Project Management

While blockchain has been widely studied in financial services (Nakamoto 2008; Yermack 2017) and supply chain systems (Saberli et al. 2019), research on its application to government project management is sparse. Most existing studies on e-governance focus on digitization and centralization (UN E-Government Survey 2020), but few investigate decentralized blockchain-based models for monitoring large-scale public projects.

2. Lack of Transparency and Accountability in Current Frameworks

[1] *Traditional government monitoring systems rely on centralized authorities, which makes them prone to manipulation, delays, and inefficiencies.* Although some digital dashboards exist for public project tracking, they often lack real-time data synchronization and tamper-proof mechanisms. There is limited research on how blockchain can embed trust and accountability by design in government project ecosystems.

3. Insufficient Use of Smart Contracts for Process Automation

Applications of smart contracts have been researched in the context of the construction sector (Das et al. 2020; Ahmadisheykhsarmast & Sonmez 2020) and healthcare systems (Mackey et al. 2019), however, the application of smart contracts for automating government project-related procedures, such as milestone payments based on project milestones, contractor performance analysis, and government compliance analysis, has hardly been investigated. This leaves a research gap for the comprehensive utilization of the potential of the blockchain technology for automating the governance.

4. Weak Citizen Engagement in Blockchain Governance Research

Research done on blockchain in governance has specifically concentrated on back-end efficiency and integrity of information (Li et al. 2019; Chaveesuk et al. 2020). Nevertheless, few studies have examined the use of blockchain technology in direct civic engagement through offering citizens transparent and verifiable information on project implementation. Research incorporating public participation in blockchain-based monitoring tools is minimal despite its significance for public accountability in governance.

5. Lack of Empirical Prototypes and Validation Studies

There have been a large number of research studies that have attempted to develop a conceptual framework for the adoption of blockchain technology in the context of governance. The lack of pilot studies on the use of blockchain technology affects its feasibility and cost effectiveness in the context of tracking public projects.

1.4 PROBLEM DEFINITION AND SCOPE

Problem Definition

The challenges in the execution and administration of government and public infrastructure projects arise due to the involvement of several entities as stakeholders, the approval channels, and the compliance regulations that are required. The involved entities of the plans, the executing agencies, the contractors, the auditors, and the financial agencies are working on different and centrally managed systems with disconnected storage of the same data, thereby resulting in disparate project reports.

Traditional project management systems are predominantly dependent on manual and sequential processes involving approvals, inspections, and funds settlements. Manual processing is often vulnerable to human mistakes, tampering, and delays, often resulting in disputes related to the completion of tasks or approval of payments. In centralized systems, there would be a lack of immutable audit trails, and accountability or

genuineness of past records would often be hard to establish due to weak compliance with role-based access control policies, thus posing threats to system security.

The need for these third-party facilitators also creates a single point of failure, thereby hampering the resilience of the system. Moreover, within a context where public finances, as well as big projects, are concerned, such shortcomings can hamper integrity, accountability, and governance. It is pertinent to acknowledge that the basic challenge that the proposed project aims to solve is the lack of a secure, transparent, and automatic system for governing multi-stakeholders of a project, whereby integrity, accountability, controlled access, and real-time visibility are ensured.

Scope of the Project

The area of focus that the project will cover includes the design and development of a permissioned blockchain solution for governance in a project on Hyperledger Fabric that will oversee the life cycle of various government and public projects. The design will be that of a consortium with eight participating stakeholder organizations that will all be identities in the blockchain network. It will implement a three-way channel design that will provide logical separation of Planning/Execution and Financial paths in order to provide confidentiality and data sharing as appropriate.

In this context, the project also involves the setup of Membership Service Providers (MSPs) for managing identity and role-based access control (RBAC) for securing user actions by organizational roles and responsibilities. The development of smart contracts (chaincode) has also been done for automating critical procedures like project enrollment, approval process, task completion submissions, work certification, compliance check, and payment authorization on certain conditions. This helps save time and prevents any disputes over imprecise and tampered data.

The project is also involved in the development of the real-time web-based user interface. The interface is equipped with role-based dashboards that showcase real-time project updates and approvals. The interface acts as an abstraction layer. It facilitates the

engagement of users who are not tech-savvy in an effective and secure manner with the blockchain.

However, it must be noted that its usage is strictly within a prototype level solution workable in an academic setting, rather than other applications such as scaling for production usage, integrating it for money transactions with other banking interfaces, and conformity to country-specific regulations and laws. Future work would certainly constitute scalability testing, interfacing for usage in legacy government interfaces, and analytics.

1.5 ASSUMPTIONS AND CONSTRAINTS

Sno.	Assumptions
1.	All participating stakeholders are verified, authorized organizations that agree to operate within a permissioned blockchain consortium and comply with predefined governance policies.
2.	Each stakeholder organization possesses the necessary technical infrastructure, including computing resources and stable network connectivity, to host blockchain nodes and access the system.
3.	Digital identities issued through Membership Service Providers are securely managed and used only by authorized personnel, ensuring reliable authentication and authorization.
4.	Project workflows, approval hierarchies, and compliance rules are clearly defined and can be accurately encoded into smart contracts without ambiguity.
5.	Data entered into the system, such as progress updates, inspection results, and compliance information, is assumed to be accurate and submitted by responsible and authorized users.
6.	External processes, including physical inspections and regulatory verifications, are assumed to be conducted correctly, with only verified outcomes recorded on the blockchain.
7.	Users access the system through the web interface using secure credentials and do not attempt to bypass role-based access control mechanisms.

TABLE 1.5.1: Assumptions

Sno.	Constraints
1.	The system is implemented as a prototype-level solution and is not evaluated under large-scale, real-world production workloads.

2.	The blockchain network is deployed with a limited number of nodes, which restricts performance, scalability, and fault-tolerance testing.
3.	Integration with external banking systems, payment gateways, or government financial platforms is not included; financial transactions are simulated through smart contract logic.
4.	The system does not fully address legal, regulatory, or jurisdiction-specific compliance requirements, which may be necessary for real-world deployment
5.	Blockchain-related factors such as transaction latency and throughput limitations impose performance constraints compared to centralized systems.
6.	The effectiveness of role-based access control and workflow enforcement is dependent on correct configuration of roles, policies, and endorsement rules.
7.	The system relies on manual data inputs for certain processes, such as progress reporting and inspections, which may affect automation completeness.

TABLE 1.5.2: Constraints

1.6 STANDARDS

System design meets industry, blockchain, and software development standards to ensure interoperability, security, reliability, and scalability. The blockchain network architecture follows industry standards as specified in Hyperledger Fabric guidelines, such as permissioned membership, channels, endorsement, and smart contract management. This will ensure that secure identity verification, access management, and consistent business transactions take place between various business entities.

Regarding identity management and authentication, the project adheres to the public key infrastructure (PKI) standard and employs the use of digital certificates issued by the certificate authority under membership service providers. This promotes secured communications, authentications, and non-repudiation for participants connected to the network. The system incorporates the different access control mechanics for roles as advocated by the standard role-based access control (RBAC) mechanism.

The smart contracts, or chaincode, follow best practices in secure code design and deterministic execution, which enables deterministic execution of business logic. Storage and processing of transaction data follow best practices in blockchain data integrity, maintaining immutability, traceability, and logging.

The web-based user interface and application layer are implemented following good software development principles such as modularity and the principle of separating concerns. Basic security on the web including authentication and secure data transfer is followed. The user interactions with the blockchain network are thus made secure.

On the whole, by following such standards, one can ensure that the proposed system will remain secure, scalable on a conceptual level, interoperable within an enterprise blockchain setting, and ready to be used in a government-oriented application, yet remain relevant in an academic field as well as in an industry.

1.7 OBJECTIVES

- Building a Hyperledger Fabric Network: Implement a secure network for 8 stakeholders to conduct business activities, comprising Planning, Execution, and Financial steps, through a 3-channel structure.
- Implement Role-Based Access Control: The MSP can be configured for providing identity management services based on role-based access control, ensuring proper authorization by using multi-level approval processes.
- Automating Processes with Chaincode: Develop smart contracts to automate the project life cycle, work certification, and condition payment processing.
- Create Real-Time UI Components: Create role-specific dashboards that display real-time progress, real-time notifications, and real-time updates on status for each role.

1.8 METHODOLOGY

Phase 1: Requirement Analysis and System Planning

The first stage is concerned with analyzing existing systems for project management and e-governance to see where there are gaps in transparency, accountability, and automation in work flow. Roles, hierarchies, and dependencies in work flow pertaining to planning, execution, and finances were analyzed. This has enabled them to identify functional and

non-functional requirements based on their findings, including matters concerning security, access, audit trails, and real-time visibility.

Phase 2: Blockchain Platform Selection

Permissioned Blockchain Framework was chosen because of enterprise requirements. Hyperledger Fabric is chosen for its modular structure, private channels, identity-based access controls, and smart contracts. Its ability to support multi-organization settings made it perfect for building a consortium model for governing.

Phase 3: Network Architecture Design

A blockchain system was designed for multi organizations, involving a total of eight stakeholder organizations. The design of the blockchain system involves peer nodes for the organizations, an ordering service for the ordering of transactions, and Certificate Authorities for identity generation. A three-channel architecture was used to segregate the Planning, Execution, and Financial processes and maintain the confidentiality of the data.

Phase 4: Identity Management and Access Control

Membership Service Providers were set up within each organization to manage digital identities. Role-based access control policies were established that controlled user actions according to organizational roles. Endorsement policies, as well as access policies, were enabled to require multi-level approvals for approved transactions like milestone certification and payment authorization.

Phase 5: Smart Contract (Chaincode) Development

The smart contracts were developed for each of the channels to implement the business logic automatically and to implement the rules of the workflow. The smart contracts are responsible for handling the project registration, approval workflow, updates, compliance checks, work certification, and the payment based on the conditions. The deterministic logic was implemented for the consistency of the workflow on all the peer nodes.

Phase 6: Application Layer and User Interface Development

A web application has been developed to communicate with the blockchain network using secure APIs. Role-specific dashboards have been designed to provide real-time status updates for projects, approvals, notifications, and audit trails. The application layer hides the complexities of blockchain technology to allow non-technical personnel to efficiently communicate with the system.

Phase 7: Testing and Validation

In order to test its effectiveness, scenarios for simulated projects have been used to test the system for access control, workflow automation, integrity within transactions, and immutability in data. Testing on functionalities ensured that smart contracts worked correctly in their tasks, while role-testing helped ensure users could not act outside their roles.

1.9 PROJECT OUTCOMES AND DELIVERABLES

Project Outcomes

This project has been able to show, in a real-life setting, the applicability of a permissioned blockchain governance system on Hyperledger Fabric. A consortium blockchain network has been developed that supports various authorized participants, featuring identity management, role-based access control, and channel isolation for the multi-channel system. This system tests the applicability of blockchain technology to deal with intricate planning, execution, and financial activities while preserving the integrity, confidentiality, and auditability of digital information. Smart contracts optimize fundamental activities associated with the project life cycle, such as approval, compliance, work certification, and authorized payments.

One of the most important results emerging through this project is improvement in the level of transparency and accountability in multi-stakeholder project implementation. In fact, because blockchain is immutable, all transactions take place in such a way that they provide a permanent record to auditors, hence making it impossible to make alterations to such transactions. Permissioned roles, in addition to multi-level approvals, ensure accountability in transactions since every action can be linked to its corresponding role.

The fact that a role-specific web-based user interface has been developed for a real-time system is a major improvement in usability and acceptability. The stakeholders have gained access to a live status update of projects, approvals, compliance, and financial authorizations in respect of their projects through simple-to-use user interfaces. The complexity of a blockchain system has been abstracted at the application level, allowing non-technical users to efficiently use the system, thus proving the applicability of blockchain technology in a governance-oriented setup.

Project Deliverables

It provides a totally configured Hyperledger Fabric blockchain ecosystem that includes several organizations representing stakeholders, peer members, ordering services, Certificate Authorities, and channels that work on three Planning, Execution, and Financial channels. This provides the basic framework on which the governance model will be implemented.

Chaincode, or smart contracts specific to each channel, will be developed to codify the business rules related to the registration process and approval progress as well as the certification and payment authorization that are based on the condition. The modules are responsible for the deterministic execution and enforcement of the workflow rules on the SO.

It comes with a full identity management solution provided by the Membership Service Providers, along with digital certificates. Finally, there are policies for role-based access control to enable safe participation.

A secured online application has been developed to allow engagement with the blockchain network. The online application provides role-specific dashboards with real-time tracking, notifications, and approvals, as well as immutable audit logs to make it easy for users to navigate the system.

1.10 NOVELTY OF WORK

The uniqueness associated with this project emerges from its use of the permissioned blockchain framework for managing the entire cycle associated with public infrastructure projects. This is beyond the traditional use associated with blockchains for storing records and/or financial dealings. This particular project differs from the traditional project management tools and the present use of blockchains by adopting a multi-channel system. This system distinguishes Planning, Execution, and Financials within the same blockchain environment. This performs the task of maintaining confidentiality and transparency.

One of the most important novel aspects of the proposed system is the use of role-based access control and multi-level approval systems. Such systems are achieved through the use of Membership Service Providers and endorsement policies. This approach ensures accountability at every step of the way throughout the entire life cycle of the project, as every single step is associated with a validated organizational identity. This approach to management further reduces reliance on central management. The use of smart contracts for automated certification of work and approval of payments based on conditions ensures a governed financial system that is rule-based and tamper-proof.

Secondly, the inclusion of the seamless integration of blockchain infrastructure with the role-centric, web-based user interface in real-time is an area of great importance. This facilitates people with no blockchain background to access the immutable ledger and automated process, thereby making it far more user-friendly and easy to adopt in the governance-oriented environment. Overall, the proposed framework provides a holistic solution for the blockchain technology specifically for the governance environment, featuring workflow and automated enforcement of compliance, among other features. It differs from the existing central systems and the piecemeal approach of the blockchain implementation.

REQUIREMENT ANALYSIS

2.1 LITERATURE SURVEY

2.1.1 RELATED WORK

A review of existing literature on government project management, e-governance, and blockchain applications reveals several important findings that contextualize the need for this project. Researchers consistently emphasize the role of transparency, accountability, and efficiency as critical determinants of successful governance systems. However, the literature also highlights persistent shortcomings in centralized digital systems and emerging gaps in blockchain adoption for public administration.

Studies on e-governance platforms, such as the United Nations E-Government Survey (2022), show that digitization has significantly improved service delivery, reduced paperwork, and increased administrative efficiency. Case studies from countries like Estonia and Singapore highlight that integrated digital governance systems can streamline processes and foster citizen trust. However, researchers also note that such systems remain vulnerable to data manipulation, cyberattacks, and single points of failure, since they rely heavily on centralized infrastructures.

Academic work on public administration theories confirms the prevalence of problems such as principal-agent issues, moral hazard, and information asymmetry in government project monitoring (Jensen & Meckling, 1976). Several studies argue that centralized platforms are unable to fully address these issues because they still rely on intermediaries to validate and disclose information.

Literature on blockchain applications in governance indicates promising opportunities. For example, Ølnes, Ubacht, and Janssen (2017) found that blockchain can strengthen trust in public records by offering immutability, decentralization, and transparency. The World Bank (2018) reported in its “Blockchain for Development” study that distributed ledger technologies can reduce inefficiencies in procurement and aid distribution by eliminating intermediaries. Pilot implementations in countries like Colombia (public

procurement) and United Arab Emirates (government services) demonstrate early successes in applying blockchain for transparency and efficiency.

At the same time, findings suggest that blockchain adoption in governance is still fragmented and experimental. According to Lemieux (2016), while blockchain has strong potential in recordkeeping and anti-corruption measures, most applications remain in pilot stages, limited to specific sectors such as land registries, supply chains, or voting systems. This aligns with Janssen et al. (2020), who argue that scalability, interoperability, and policy alignment are key challenges that have slowed widespread blockchain deployment in public administration.

Furthermore, existing research highlights that citizen engagement remains limited in most digital and blockchain experiments. While governments have implemented portals for information sharing, scholars such as Bannister and Connolly (2012) argue that genuine participatory governance requires mechanisms that provide citizens with real-time, verifiable, and user-friendly access to project data, which remains an underdeveloped area in the literature.

In summary, existing literature establishes that digitization has improved efficiency but failed to eliminate corruption and inefficiencies due to centralization. Blockchain emerges as a theoretically sound and practically promising solution, with features such as immutability, decentralization, and smart contracts directly addressing these shortcomings. However, research findings also underscore that current blockchain initiatives are narrow, experimental, and lack comprehensive frameworks that integrate multiple stakeholders, cover diverse domains, and ensure citizen engagement. This creates an opportunity for the present project to advance the field by proposing a holistic blockchain-based system for government project tracking.

2.1.2 RESEARCH GAPS OF EXISTING LITERATURE

A close look at the current research on e-governance, how governments manage projects, and blockchain-based systems shows that there are several areas that still need more attention. This justifies the need for the work that is being proposed.

First, even though digital tools have made government work more efficient and services better, these systems still face some big problems.

Most of them use centralized databases, which are easy to hack, can be changed without permission, and have one main point that could fail. The research hasn't found good ways to replace these centralized systems with decentralized, secure alternatives that ensure trust and monitoring from start to finish.

Second, government projects often face issues like conflicts between leaders and workers, dishonest behavior, and unequal access to information.

But current studies haven't explored how new technology can help make these processes more accountable. There's not much research on how to use automatic checks and unchangeable records to fix these long-standing problems.

Third, even though blockchain is seen as a great way to increase transparency and trust, it's still being used in very few and small ways in public government work.

Most uses are in specific areas like land records, buying goods, or voting. There's a lack of complete blockchain systems that can handle the whole life of a government project, from planning to management and financial tracking.

Another issue is that not much attention is given to involving many different groups.

Many studies on blockchain focus on making the back-end more efficient, but they don't look enough at how to bring together different groups like government departments, contractors, auditors, and citizens. There's not much research on how to manage access and workflows securely across multiple organizations.

Also, public involvement is not well developed in both traditional e-governance and blockchain projects.

Although there are websites that show information, the research points out that there are not enough easy-to-use tools that allow people to track progress and spending in real time. Studies that combine blockchain with ways for citizens to be involved are very limited.

Finally, the studies mention problems like handling large amounts of data, connecting different systems, and following rules, but they don't give much real-world testing.

A lot of the research is still just ideas, without actual working models that show how well they can be used in real government situations.

2.1.3 DETAILED PROBLEM ANALYSIS

The management and oversight of public projects face challenges due to systemic issues that affect every stage of the project lifecycle. Current digital governance platforms use centralized architectures. This reliance makes them vulnerable to unauthorized changes and selective data sharing. Centralized control prevents the creation of a single, tamper-proof source of truth and can delay transparency. Additionally, the lack of synchronized tracking for project funds and milestones creates significant administrative delays and poor decision-making. Without cryptographic verification, this lack of clarity often leads to misallocated resources, budget overruns, and project delays.

One major reason for these issues is the information gap between stakeholders. Government agencies, contractors, and auditors often work in isolation, with little data verification. This fragmented setup encourages corruption, misreporting, and fraud as money passes through multiple intermediaries. Meanwhile, citizens usually cannot monitor progress in real-time. They receive only periodic and filtered reports that may not accurately reflect the true status of public projects.

Traditional auditing and compliance practices worsen these problems because they respond to issues instead of preventing them. Since audit trails can be altered or hidden, records may be changed to cover up wrongdoing, often discovered only after significant financial or infrastructure harm has occurred. This lack of transparency has created a major trust gap between the government and its citizens, discouraging public involvement and weakening democratic accountability.

Finally, while blockchain could provide a solution, its use in governance remains scattered and mainly focuses on areas like land registries or procurement. There is a clear need for a unified framework that can track various government projects using an

integrated, decentralized, and tamper-proof solution. This gap emphasizes the urgent need for a system that ensures accountability and citizen engagement by embedding trust into the project management framework.

2.1.4 SURVEY OF TOOLS AND TECHNOLOGY USED

The monitoring and management of government projects have historically relied on a variety of tools and technologies that aim to enhance efficiency, transparency, and accountability. However, these systems differ in scope, architecture, and effectiveness. A survey of the existing landscape highlights both the progress made and the limitations that create space for blockchain-based innovation.

1. Project Management Software

Commercial project management platforms such as Microsoft Project, Oracle Primavera, and Jira are widely used for scheduling, resource allocation, and milestone tracking. These tools help project managers create detailed work breakdown structures, assign responsibilities, and monitor progress. However, their application in government projects is often restricted to internal departmental use, and they do not provide public transparency or immutable audit trails.

2. Centralized E-Governance Platforms

Governments worldwide have launched digital platforms to monitor large-scale projects. For example, India's PRAGATI platform integrates project tracking with video conferencing for inter-ministerial coordination, while the Central Public Procurement Portal (CPPP) digitizes procurement processes. Similar systems exist globally, such as the Estonian e-Governance Framework and Singapore's Smart Nation initiatives. These systems streamline monitoring but remain centralized, making them prone to data manipulation, cyberattacks, and selective reporting.

3. Enterprise Resource Planning (ERP) Systems

Large government departments and public sector organizations use ERP solutions such as SAP and Oracle ERP to integrate financial management, procurement, and project

execution processes. These systems enhance internal efficiency and provide detailed reporting capabilities. However, ERP systems are costly, complex to maintain, and remain inaccessible to citizens, thereby failing to address the need for participatory transparency.

4. Audit and Compliance Tools

Traditional auditing relies on tools for financial tracking and compliance reporting, including software like ACL Analytics and CaseWare IDEA. While such tools improve the efficiency of audits, they are largely reactive, identifying problems after they occur rather than preventing them. Moreover, audit data often depends on departmental disclosures, which can be delayed or manipulated.

5. Open Data Portals

Open data initiatives promoted by organizations such as the Open Government Partnership (OGP) encourage governments to share project-related information with the public. Countries like the United States, the United Kingdom, and India have established open data portals that provide citizens with access to datasets on spending, infrastructure, and service delivery. While these portals represent progress toward transparency, their data is not always real-time, verifiable, or comprehensive, limiting their impact on accountability.

6. Blockchain-Based Pilots

Emerging research and pilot projects have experimented with blockchain in governance. For instance, Colombia tested blockchain for public procurement, while the World Bank's Blockchain for Development initiative explored blockchain's potential in aid distribution. The United Arab Emirates launched a national blockchain strategy to digitize government services. These efforts showcase blockchain's potential in ensuring immutability and transparency but remain limited to specific domains and have not yet evolved into holistic, large-scale systems for project monitoring.

2.1.5 SUMMARY

This project builds upon the foundational success of existing e-governance initiatives and centralized monitoring platforms like PRAGATI while addressing their core architectural vulnerabilities. While previous systems successfully digitized project tracking and procurement, they remained dependent on centralized databases, which are susceptible to data manipulation, single points of failure, and selective disclosure. This work differs by replacing these centralized data management structures with a decentralized blockchain framework, ensuring that project-related transactions, milestones, and fund allocations are recorded on an immutable, single source of truth.

The proposed solution further advances beyond current fragmented blockchain pilots, which are often limited to narrow domains like land registries or procurement. Unlike these precursors, this work introduces a holistic, end-to-end monitoring system that covers the entire project lifecycle—from budget allocation and milestone-based smart contract automation to real-time citizen oversight via role-based dashboards. By embedding accountability directly into the technological core through self-executing smart contracts and immutable audit trails, the system shifts the governance model from one based on institutional trust to one based on cryptographic verification.

2.3 SOFTWARE REQUIREMENT SPECIFICATIONS

2.3.1 INTRODUCTION

2.3.1.1 PURPOSE

The main objective of this project is to develop a system for the transparent, efficient, and tamper-proof tracking of government projects on a blockchain system. Government projects are often marred with time delays, cost escalations, and resource misallocation due to corruption, accountability, and inefficient tracking systems. The traditional systems are digitized but vulnerable to fraud and do not offer adequate real-time tracking facilities for the common citizen.

This proposes to use the intrinsic characteristics of the blockchain technology, such as immutability, decentralization, and consensus, to build a platform where each of the stakeholders, including the governmental agencies, the contractors, auditors, and the citizens, will be able to interact through a common ledger of the activities performed in

the project. Smart contracts will be used to automate the process of releasing the funds and the evaluation of the milestones.

Therefore, the design of the system is not only aimed at increasing efficiency in the process of executing projects but also at regaining the trust of the citizens by allowing them to access the progress and expenditure information of the projects in a transparent and real-time manner. Finally, the last goal of the entire project is to create an environment of accountability in the system of governance, where accountability is designed in the technology used.

2.3.1.2 INTENDED AUDIENCE AND READING SUGGESTIONS

- Government authorities and administrative bodies will use the system for project registration, fund allocation, and real-time monitoring.
- Contractors and implementing agencies will interact with the system to submit milestone updates, request payments, and verify compliance via smart contracts.
- Auditors and regulatory institutions will rely on the immutable records and audit trails for independent performance evaluation and fraud detection.
- Citizens and civil society organizations will access public dashboards to track project progress, expenditures, and deadlines in real time.
- Policymakers and governance experts will use system outputs to evaluate governance reforms, policy efficiency, and citizen trust indicators.

Reading Suggestions

Readers who are new to blockchain technology are advised to begin with the introductory sections that explain fundamental concepts such as permissioned blockchains, Hyperledger Fabric, smart contracts, and role-based access control. These sections provide the necessary background to understand the system architecture and design decisions presented later in the report. Non-technical readers may focus on the problem definition, proposed solution, and outcomes to gain insights into the

governance and transparency benefits of the system without delving into implementation details.

Technical readers and practitioners are encouraged to study the methodology, system architecture, and smart contract design sections in detail, as these provide a comprehensive understanding of the implementation approach. For a complete perspective, readers may also review the assumptions, constraints, novelty of work, and future scope sections to understand the limitations and potential extensions of the proposed system. Reading the report sequentially is recommended for a holistic understanding, while specific sections can also be referenced independently based on the reader's area of interest.

2.3.1.3 PROJECT SCOPE

The scope of this project is to design and develop a blockchain-based platform that provides end-to-end monitoring of government projects. The system will encompass all major aspects of project management, including project registration, fund allocation, milestone definition, progress tracking, auditing, and citizen engagement. Government authorities will be able to register new projects, allocate budgets, and define specific timelines and deliverables. Contractors and implementing agencies will interact with the system by submitting progress updates and requesting fund releases as milestones are completed.

Smart contracts will serve as the backbone of the system, ensuring that payments are automatically released once predefined conditions are verified, thereby reducing delays and minimizing the risk of corruption. Auditors and regulatory institutions will benefit from immutable audit trails and real-time access to verifiable data, enabling independent evaluations without reliance on intermediaries. Citizens and civil society organizations will be empowered through public dashboards that display transparent, real-time information on project progress, expenditures, and deadlines, thus fostering participatory governance.

The system will support multiple stakeholders through role-based access controls, ensuring that each user interacts with the system according to their responsibilities and privileges. Built on blockchain infrastructure, the platform will guarantee decentralization, immutability, and transparency, addressing the shortcomings of traditional centralized monitoring systems. The scope also includes both backend blockchain development and the creation of user-friendly frontend applications tailored to different stakeholders. Ultimately, the project aims to reduce corruption, eliminate inefficiencies, strengthen accountability, and rebuild public trust in the management of government-funded initiatives.

2.3.2 OVERALL DESCRIPTION

2.3.2.1 PRODUCT PERSPECTIVE

The proposed blockchain-based government project tracking system is envisioned as an innovative solution that integrates seamlessly into the broader ecosystem of e-governance platforms. While many existing systems for project monitoring are centralized and operate within departmental silos, this product is designed as a decentralized and transparent alternative that directly addresses the inefficiencies and trust deficits associated with traditional approaches.

From a structural perspective, the system will function as a distributed ledger platform where all stakeholders—government authorities, contractors, auditors, and citizens—interact with a common source of truth. Unlike conventional project management tools that depend on intermediaries for verification, this product eliminates the need for manual validations by using smart contracts to automate fund disbursements, milestone approvals, and compliance checks.

The product will also serve as a complementary extension to existing e-governance frameworks, procurement portals, and ERP systems. For example, project registration and financial allocation data from current systems can be integrated into the blockchain platform to create a unified, verifiable audit trail. The public-facing dashboards will further extend the scope of existing transparency portals by providing real-time, verifiable data rather than static or delayed reports.

In terms of design philosophy, the system is modular, comprising three key layers:

1. **Blockchain Layer** for decentralized storage, consensus management, and immutability.
2. **Application Layer** for smart contract execution, milestone tracking, and stakeholder interactions.
3. **Interface Layer** providing customized dashboards and user interfaces for each stakeholder group.

The product perspective thus positions the system not as a standalone application but as a next-generation governance tool that enhances existing mechanisms by embedding accountability, transparency, and automation into the very fabric of project execution. By reimagining how data is recorded, verified, and shared, the product establishes a robust framework for participatory governance, efficient resource management, and long-term public trust.

2.3.2.2 PRODUCT FEATURES

The blockchain-based government project tracking system is designed with a comprehensive set of features that address the shortcomings of existing centralized monitoring platforms while promoting transparency, accountability, and efficiency. The following are the core features of the system:

Project Registration and Initialization

Government authorities can register new projects on the system by providing details such as project title, description, budget allocation, timelines, and implementing agencies. Once registered, the project becomes a permanent entry on the blockchain, ensuring that its details cannot be tampered with.

Smart Contract Automation

The system employs smart contracts to govern project milestones and fund disbursements. Funds are automatically released once predefined conditions, such as the

successful completion of a milestone verified by stakeholders, are met. This eliminates bureaucratic delays and reduces opportunities for corruption.

Milestone Tracking and Progress Updates

Contractors and implementing agencies can submit real-time updates on project milestones, including completion status, expenditure reports, and supporting documents. These updates are stored on the blockchain, creating a transparent and immutable project timeline.

Immutable Audit Trails

Every transaction, update, or modification related to a project is recorded on the blockchain, forming an immutable audit trail. Auditors and oversight bodies can independently verify the accuracy of records without relying on manually curated reports, thereby strengthening accountability.

Role-Based Access Control

Different stakeholders interact with the system according to their roles. For example, government officials can allocate funds, contractors can update milestones, auditors can review records, and citizens can access transparent dashboards. This ensures security while maintaining inclusivity.

Public Transparency Dashboards

Citizens and civil society organizations have access to dashboards that display project progress, financial utilization, and timelines in real time. These dashboards simplify complex project data into accessible visual insights, fostering participatory governance.

Interoperability with Existing Systems

The system is designed to integrate with existing government e-governance platforms, procurement portals, and ERP solutions. This interoperability ensures seamless data flow while avoiding duplication of efforts and resources.

Security and Data Integrity

Built on blockchain infrastructure, the system guarantees decentralization, immutability, and resistance to tampering. Data encryption and consensus mechanisms ensure that only valid transactions are added to the ledger, maintaining trust across stakeholders.

Scalability and Flexibility

The platform is designed to accommodate projects of varying sizes, from small local initiatives to large-scale national infrastructure programs. Modular architecture allows the system to be extended with additional features or adapted to different governance contexts.

2.3.3 EXTERNAL INTERFACE REQUIREMENTS

2.3.3.1 USER INTERFACES

- Banking System Interface: Payment processing, fund transfers
- Government API Integration: Citizen verification, project registration
- Blockchain Network Interface: Smart contract deployment, transaction processing
- Notification Services: Email/SMS providers, push notification services
- Document Management: File upload/download, document versioning

2.3.3.2 SOFTWARE INTERFACES

- Government Authority Portal: Project creation, funding approval interfaces
- Contractor Portal: Milestone submission, progress reporting
- Auditor Dashboard: Compliance checking, audit trail viewing
- Citizen Dashboard: Public project viewing, transparency features

2.3.4 OTHER NON-FUNCTIONAL REQUIREMENTS

2.3.4.1 PERFORMANCE REQUIREMENTS

The system shall provide reliable and responsive performance while supporting multiple stakeholders accessing the platform concurrently. User interactions such as dashboard viewing, project tracking, and audit log retrieval must execute within acceptable response times.

Smart contract transactions, including project registration, milestone submission, approvals, and payment authorization, shall be processed efficiently with minimal latency

suitable for a permissioned blockchain environment. The system must handle moderate transaction throughput without performance degradation.

Read operations shall support near real-time visibility of project data, while maintaining the integrity and immutability of blockchain records. During peak usage, the system should queue and process transactions securely without data loss or failure.

The architecture shall support future performance optimization and scalability without requiring major structural changes.

2.3.4.2 SAFETY REQUIREMENTS

- The system must ensure that no single point of failure disrupts government project tracking, by implementing redundant nodes and backup mechanisms.
- All smart contracts must be rigorously tested and verified to prevent execution errors that could halt project operations or misallocate funds.
- The system must incorporate graceful degradation so that essential services (e.g., citizen dashboards, audit logs) remain accessible even during partial failures.
- Proper monitoring and alerting mechanisms should be in place to notify administrators of anomalies, hardware crashes, or performance bottlenecks.
- The safety framework must comply with government IT regulations and standards, ensuring that the platform is resilient to both technical and operational risks.

2.3.4.3 SECURITY REQUIREMENTS

The system incorporates a multi-layered security framework designed to protect sensitive data and ensure the absolute integrity of the decentralized ledger.

- **Immutable Audit Trails:** Every transaction, update, or modification related to a project is recorded on the blockchain to form an unchangeable history, ensuring that data cannot be altered or hidden to conceal misconduct.
- **Role-Based Access Control (RBAC):** Identity management is enforced through Membership Service Providers (MSPs) and Decentralized Identifiers (DIDs) to

ensure stakeholders only interact with the system according to their specific responsibilities.

- **Data Encryption and Integrity:** The architecture utilizes encryption techniques for sensitive financial data and employs consensus mechanisms to ensure only valid, verified transactions are added to the ledger.
- **Smart Contract Security:** All chaincode must undergo rigorous testing and formal verification to minimize the risk of coding errors, vulnerabilities, or unintended execution that could misallocate funds.
- **Communication Security:** TLS/SSL protocols are implemented to safeguard all data transmissions between user interfaces, APIs, and the network nodes.
- **Endpoint Protection:** The framework includes vulnerability testing and monitoring to safeguard against cyberattacks at data entry interfaces and user devices.

2.4 RISK ANALYSIS

The development and deployment of a blockchain-based governance system for public infrastructure projects involve several technical, operational, and organizational risks that must be identified and addressed to ensure system effectiveness and reliability. This section analyzes the potential risks associated with the proposed solution and highlights their possible impact on project execution.

One of the primary risks is technical complexity, as blockchain platforms such as Hyperledger Fabric require specialized knowledge for network configuration, smart contract development, and identity management. Improper configuration of channels, endorsement policies, or access control mechanisms may lead to security vulnerabilities or system malfunctions. Additionally, errors in smart contract logic can result in incorrect workflow automation or unintended authorization of transactions, affecting project governance.

Another significant risk is scalability and performance limitation. As the number of stakeholders, transactions, and project records increases, the blockchain network may experience increased latency and reduced throughput compared to centralized systems.

Although the proposed system is designed as a prototype, performance constraints could impact real-world adoption if not adequately addressed through optimization and scaling strategies.

Data accuracy and input dependency present further risks, as the correctness of blockchain records depends on the quality of data submitted by authorized users. While the blockchain ensures immutability, it cannot inherently validate the accuracy of external inputs such as inspection reports or progress updates. Inaccurate or biased data entry may lead to incorrect automated decisions and financial authorizations.

The project also faces organizational and adoption risks, particularly resistance to change from stakeholders accustomed to traditional centralized systems. Successful operation of the proposed solution requires stakeholder cooperation, policy alignment, and adequate training. Lack of user acceptance or improper role assignment can reduce system effectiveness and limit transparency benefits.

Security and identity management risks must also be considered. Although cryptographic identities enhance security, compromise or misuse of digital certificates may allow unauthorized access or actions. Additionally, mismanagement of private keys or credentials could undermine access control mechanisms and accountability.

Finally, regulatory and legal risks exist due to the evolving nature of blockchain governance frameworks. The absence of clearly defined legal standards for blockchain-based public project management may pose challenges for large-scale deployment. Compliance with jurisdiction-specific data protection and governance regulations would be essential in real-world implementations but is outside the scope of this academic project.

Overall, while the proposed system offers significant advantages in transparency and automation, careful consideration and mitigation of these risks are essential for practical adoption and long-term sustainability.

METHODOLOGY ADOPTED

3.1 INVESTIGATIVE TECHNIQUES

Investigative Technique Involved : **Descriptive**

The investigative technique used in this project is descriptive. The focus is on designing, developing, and analyzing a new system model for governance-oriented project management with blockchain technology. The goal is not to compare existing systems or test hypotheses with controlled variables. Instead, the aim is to observe, document, and describe how a permissioned blockchain framework can be structured and applied to tackle real-world challenges in public infrastructure project execution. The project looks at the feasibility, structure, and operational behavior of a blockchain-based governance model by designing its architecture, workflows, access control methods, and automation logic. This approach aligns with the descriptive investigative method.

The methodology starts with a thorough analysis of existing centralized project management and e-governance systems. This helps identify issues related to transparency, accountability, and workflow inefficiencies. Based on these observations, we defined system requirements and designed a conceptual model to fill the gaps. The descriptive approach is evident in how the project documents stakeholder roles, approval processes, and workflow dependencies across planning, execution, and finance without trying to change these factors.

Next, we designed and implemented a permissioned blockchain architecture using Hyperledger Fabric. We configured the network structure, including organizations, peer nodes, ordering services, and channels, to show how multi-stakeholder coordination can work through decentralized but controlled data sharing. The three-channel architecture helps us observe how isolating workflows improves confidentiality and governance clarity. We implemented identity management through Membership Service Providers and role-based access control policies to create secure access enforcement methods within a real-world governance context.

Additionally, we developed smart contracts to automate activities in the project lifecycle, such as approvals, progress checks, work certification, and condition-based payment authorization. We recorded the execution of these smart contracts to show how automation cuts down on manual work and improves process integrity. After that, we created a real-time web-based user interface to visualize project data, approvals, and audit logs. This allows for descriptive analysis of system usability and real-time transparency.

Finally, we validated the system using simulated project scenarios and documented observations related to transparency, traceability, access control effectiveness, and workflow efficiency. These observations help evaluate the proposed system's effectiveness. Therefore, the project follows a descriptive investigative technique by systematically designing, implementing, and documenting a new blockchain-based governance framework and analyzing its behavior and outcomes in a structured way.

3.2 PROPOSED SOLUTION

The proposed solution is a permissioned blockchain-based platform for managing government and public infrastructure projects. It securely and transparently handles the entire lifecycle of these projects. The system uses Hyperledger Fabric to create a consortium blockchain network that connects multiple authorized stakeholders involved in planning, executing, auditing, and managing finances. By replacing scattered and centralized project management systems with a decentralized, yet controlled ledger, the solution ensures data integrity, traceability, and accountability throughout all project stages.

To address confidentiality and workflow complexity, the system uses a three-channel structure that separates Planning, Execution, and Financial processes. Each channel has its own ledger and smart contracts, allowing stakeholders to access only the information relevant to their roles while maintaining overall system transparency. This separation prevents unnecessary data exposure and reduces inter-departmental conflicts, while still enabling coordinated project execution.

The proposed system includes strong identity management and role-based access control through Membership Service Providers. Each stakeholder organization receives cryptographic identities, and users are assigned specific roles like planning authority, contractor, auditor, or finance officer. Access permissions and endorsement policies require multi-level approvals for important actions, ensuring no single entity can change project records or make financial decisions independently.

Smart contracts, or chaincode, drive workflow automation in this proposal. These contracts outline the business rules for project registration, milestone approvals, progress checks, compliance validation, work certification, and conditional payment authorization. Once certain conditions are met, smart contracts automatically execute, reducing manual input, eliminating confusion, and ensuring consistent enforcement of governance rules.

To make the solution user-friendly, it features a real-time web-based interface that allows stakeholders to interact with the blockchain network through secure APIs. Role-specific dashboards provide live updates on project progress, approval status, compliance indicators, financial authorizations, and immutable audit logs. Automated notifications and alerts enhance coordination and support timely decision-making.

In summary, the proposed solution offers a complete governance framework that combines decentralization, controlled access, automation, and real-time transparency. By ensuring tamper-proof records, automatic compliance enforcement, and improved coordination among stakeholders, the system addresses key weaknesses of traditional project management methods and establishes a scalable foundation for clear and accountable public project execution.

3.3 WORK BREAKDOWN STRUCTURE

The Work Breakdown Structure (WBS) for this project decomposes the overall blockchain-based governance system into manageable and logically sequenced work packages. The WBS is designed to ensure systematic planning, efficient execution, and clear responsibility allocation throughout the project lifecycle. The project is divided into

major phases, each consisting of well-defined activities that collectively lead to the successful development and validation of the proposed system.

Project Initiation and Requirement Analysis

This phase focuses on understanding the project area and determining the need for a blockchain-based governance solution. Activities include studying current public project management systems, examining governance issues, identifying stakeholders, and defining the project scope. We derive functional and non-functional requirements based on needs for transparency, security, access control, and automation. The result of this phase is a clear definition of the problem and a specification of requirements.

System Design and Architecture Planning

In this phase, the overall system architecture is designed. This includes selecting Hyperledger Fabric as the blockchain platform, defining the consortium network structure, and planning the three-channel architecture for Planning, Execution, and Financial workflows. Identity management using Membership Service Providers and role-based access control policies are also designed. This phase results in detailed architectural and workflow designs.

Blockchain Network Setup

This phase focuses on setting up the blockchain infrastructure. Activities include configuring peer nodes, ordering services, certificate authorities, and channels for different stakeholder organizations. We define and implement network policies, endorsement rules, and access permissions. Finishing this phase successfully results in a working permissioned blockchain environment.

Smart Contract (Chaincode) Development

In this phase, smart contracts are developed for each channel to automate project lifecycle processes. Activities include coding chaincode for project registration, approval workflows, progress tracking, compliance validation, work certification, and

condition-based payment authorization. The smart contracts are tested for deterministic execution and correctness.

Application Layer and User Interface Development

This phase involves developing a web-based application to interact with the blockchain network. Role-specific dashboards are created for different stakeholders, displaying real-time project status, approvals, notifications, and audit logs. Secure APIs are implemented to connect the user interface with the blockchain backend.

System Testing and Validation

Testing and validation activities are conducted to ensure correct system behavior. This includes functional testing of smart contracts, validation of role-based access control, verification of transaction immutability, and testing of workflow automation using simulated project scenarios. Issues identified during testing are resolved iteratively.

Documentation and Reporting

This phase includes preparing technical documentation covering system architecture, smart contracts, workflows, and configuration details. Academic documentation such as methodology, results, risk analysis, and future scope is also compiled. This phase ensures knowledge transfer and evaluation readiness.

Final Review and Project Submission

The final phase involves reviewing the complete system and documentation, incorporating feedback from mentors or evaluators, and preparing the final project submission. This phase concludes with demonstration readiness and evaluation preparation.

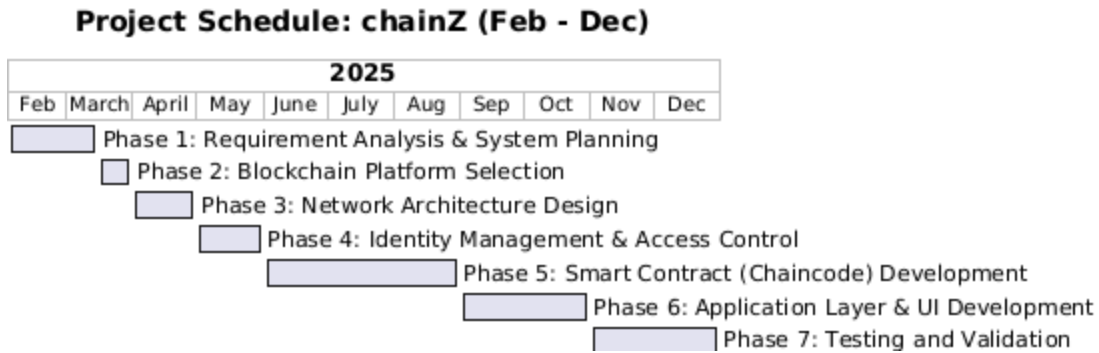


Fig 3.3.1 Gantt Chart

3.4 TOOLS AND TECHNOLOGY USED

The development of the blockchain-based government project tracking system required a combination of software tools, programming environments, and supporting technologies. Each component was carefully selected to ensure compatibility, scalability, and ease of deployment within the scope of a capstone project.

Blockchain Platform

- Hyperledger was chosen as the blockchain platform due to its mature ecosystem and support for smart contracts.
- Ganache was used to simulate a local Ethereum blockchain for development and testing purposes.
- Truffle Suite facilitated smart contract compilation, deployment, and testing.

Smart Contract Development

Solidity served as the primary programming language for developing smart contracts that handle project registration, milestone verification, and automated fund release.

Frontend Development

- React.js was used to create interactive user interfaces for various groups such as government officials, contractors and regular people.
- Web3.js library helped connect the React frontend and Ethereum blockchain.

Development Tools

- Visual Studio Code (VS Code) as the integrated development environment.
- GitHub for version control and collaborative development.
- Postman for testing APIs.
- Deployment and Hosting
- Netlify for hosting the frontend interface.
- Infura to provide scalable Ethereum node access without requiring local full-node maintenance.
- Testing and Security
- Mocha/Chai for unit testing smart contracts.
- MythX (or Remix IDE security plugins) for smart contract security analysis.

DESIGN SPECIFICATIONS

4.1 SYSTEM ARCHITECTURE

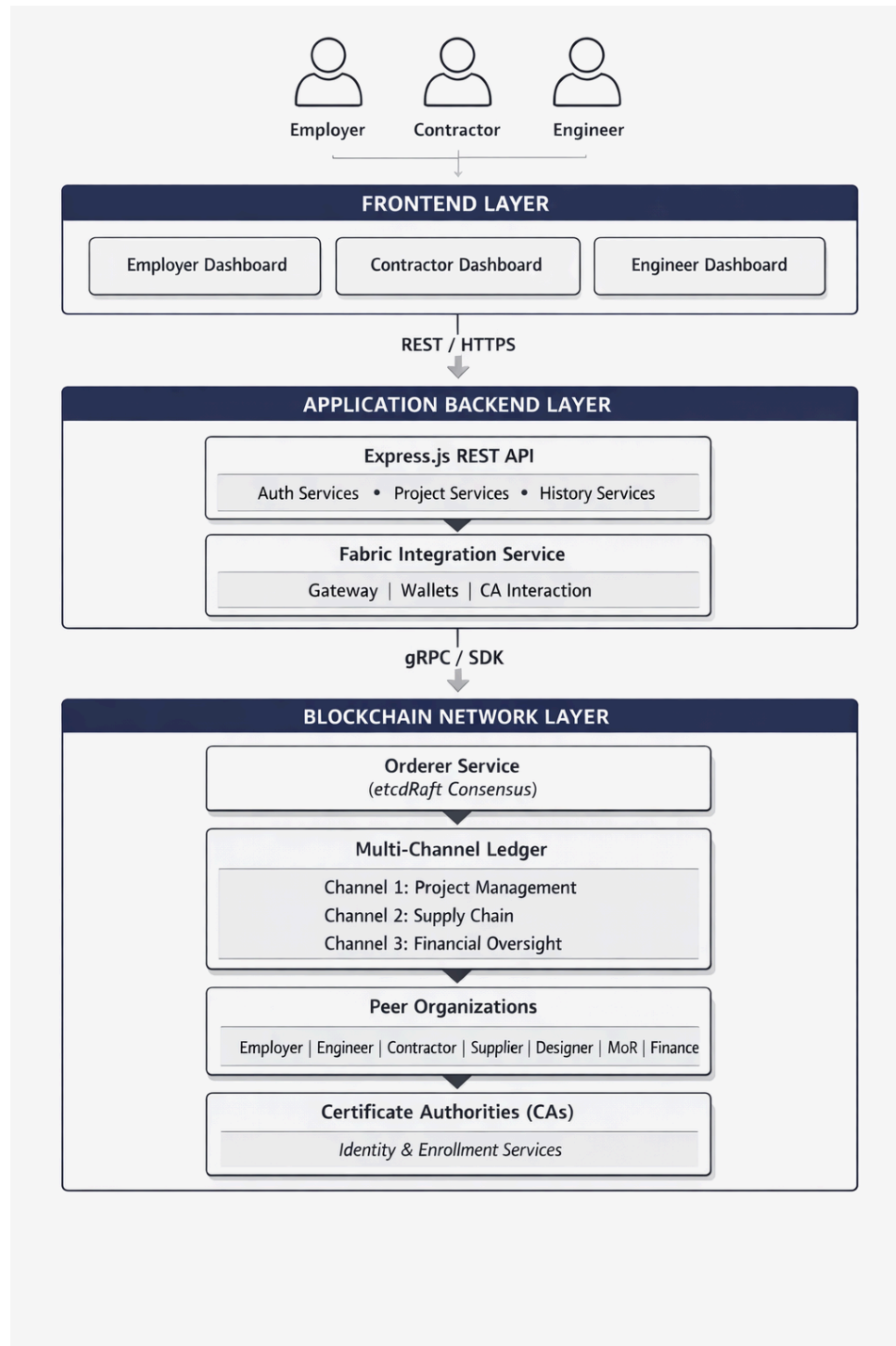


Fig 4.1.1 System architecture

The proposed blockchain-based government project tracking system uses a layered architecture to guarantee modularity, scalability, and security. Each layer has a specific

set of functions and collectively helps to monitor the projects in a transparent and tamper-proof way.

1. Security Layer

This layer makes sure all communication and data are secure. It uses TLS/SSL to protect data sent between people and the system, and encryption to keep sensitive project info, financial details, and audit logs safe. Since the system deals with important government contracts and money, security is built in from the start.

2. Presentation Layer

This is where users interact with the system. It includes a Web Interface for end-users and an API Gateway that allows safe communication with other apps. This layer ensures the system is easy to use, offers different dashboards for various roles, and is accessible to all users.

3. Service Layer

The service layer includes the core services, each responsible for a crucial function:

- Auth Service: Handles authentication, access control, and user identity management.
- Project Service: Manages project registration, tracking milestones, and updates.
- Fund Service: Takes care of fund distribution, milestone-based disbursement, and payment records.
- Audit Service: Maintains immutable audit trails, ensuring everything is done properly and everyone is held accountable.

4. Business Logic Layer

This layer handles the operational logic of the system, aligning technical processes with governance rules and policies. It contains:

- Workflow Engine: Controls how projects get approved, milestones are checked and money is released.
- Business Rules: Encodes rules and policies set by the government and specific to projects so they are baked into the system
- Notifications: Sends real-time updates to stakeholders about the project progress, fund transfers, or any issues.

5. Blockchain Layer

The blockchain layer ensures immutability, transparency, and decentralization of project records. It consists of:

- Smart Contracts: Automatically release money and check compliance based on set rules.
- Consensus Mechanism: Checks that all transactions are valid and trusted across the network.
- Chaincode: Holds the instructions that run on the blockchain.

6. Data Layer

This layer is responsible for managing the persistence of data. It integrates both blockchain and off-chain storage:

- Blockchain Ledger: Stores permanent records of project transactions, smart contract runs, and audits.
- Off-Chain Database: Holds large files such as project reports, blueprints, or multimedia evidence like drawings or videos.
- Cache: Helps speed up access to frequently used data.

7. External Systems

The system works with outside services to connect with existing government and financial tools:

- Banking API: For secure fund transfers and validation of transactions.

- Gov API: Integrates national ID systems, procurement sites, or other e-governance services.
- Email/SMS: Sends messages and alerts to users about project updates and other critical information.

4.2 DESIGN LEVEL DIAGRAMS

DFD level 0

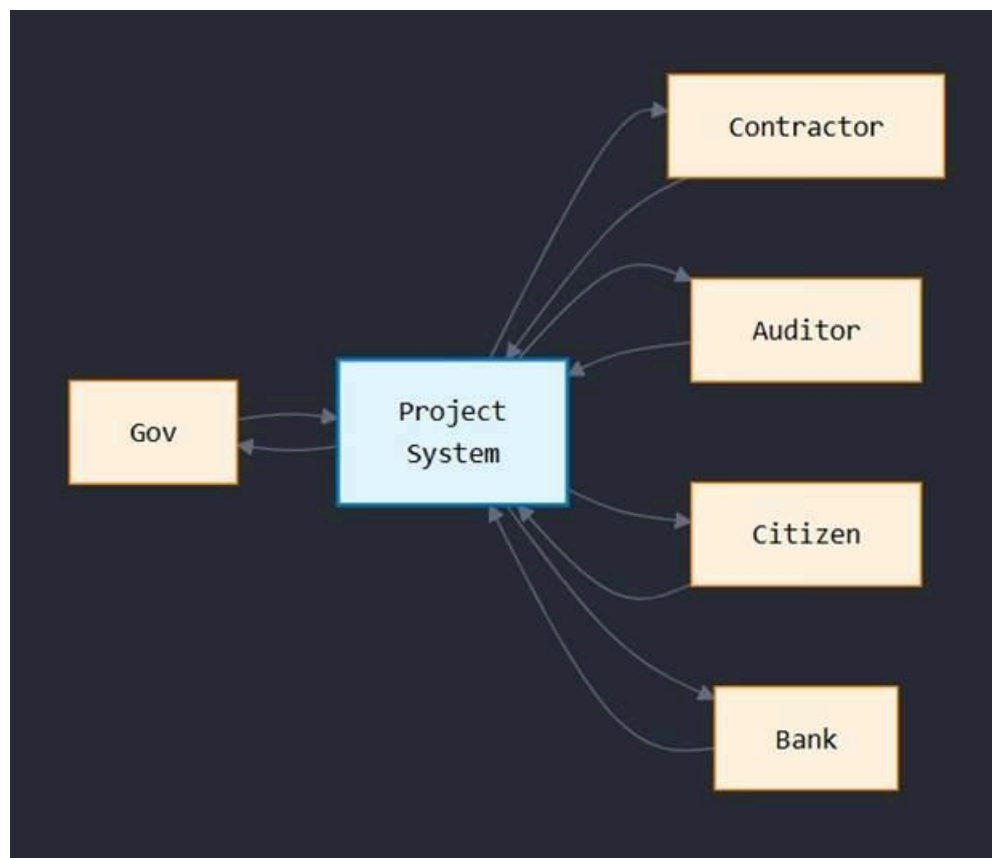


Fig 4.2.1 DFD Level 0

DFD level 1

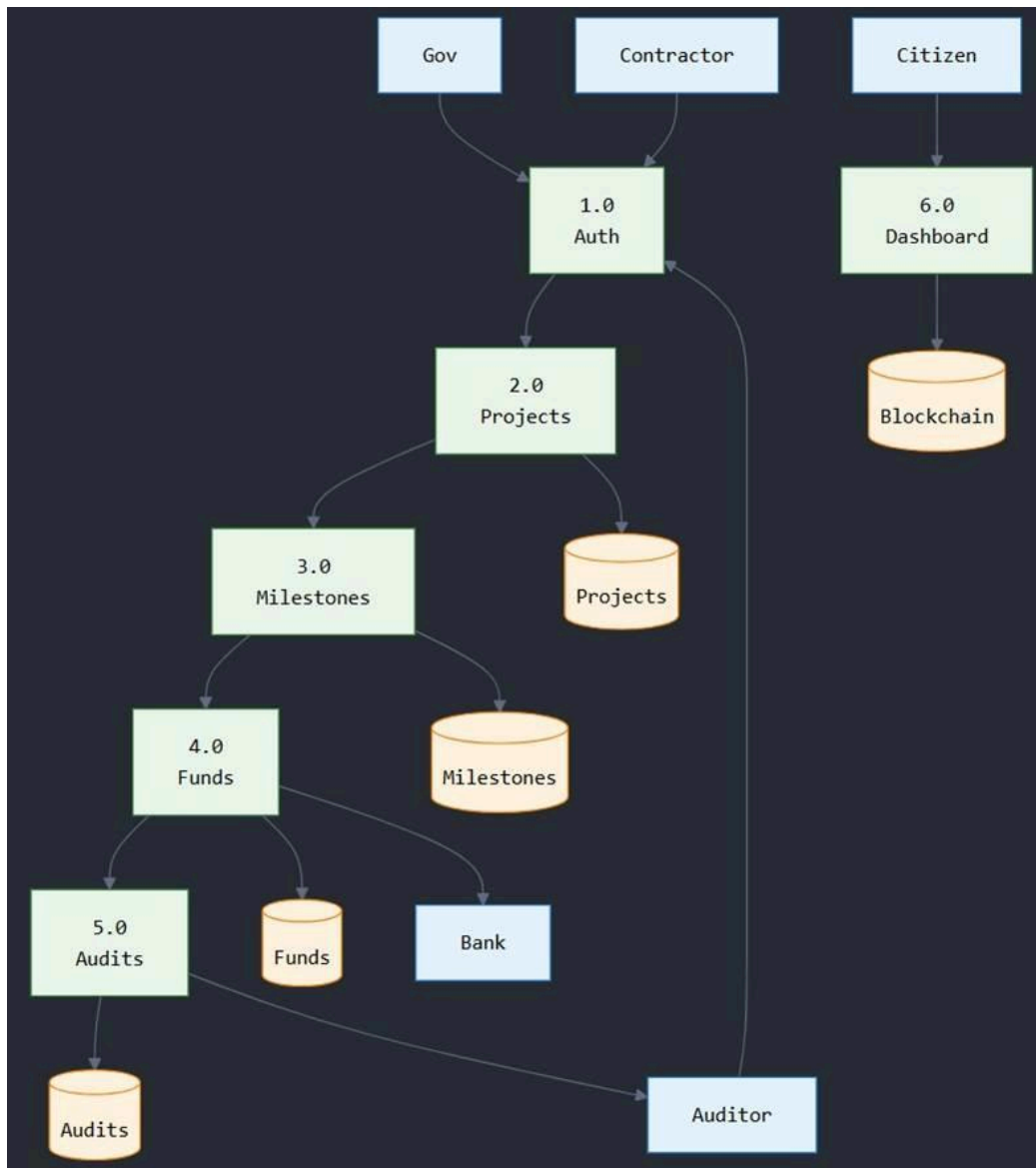


Fig 4.2.2 DFD Level 1

Sequence Diagram

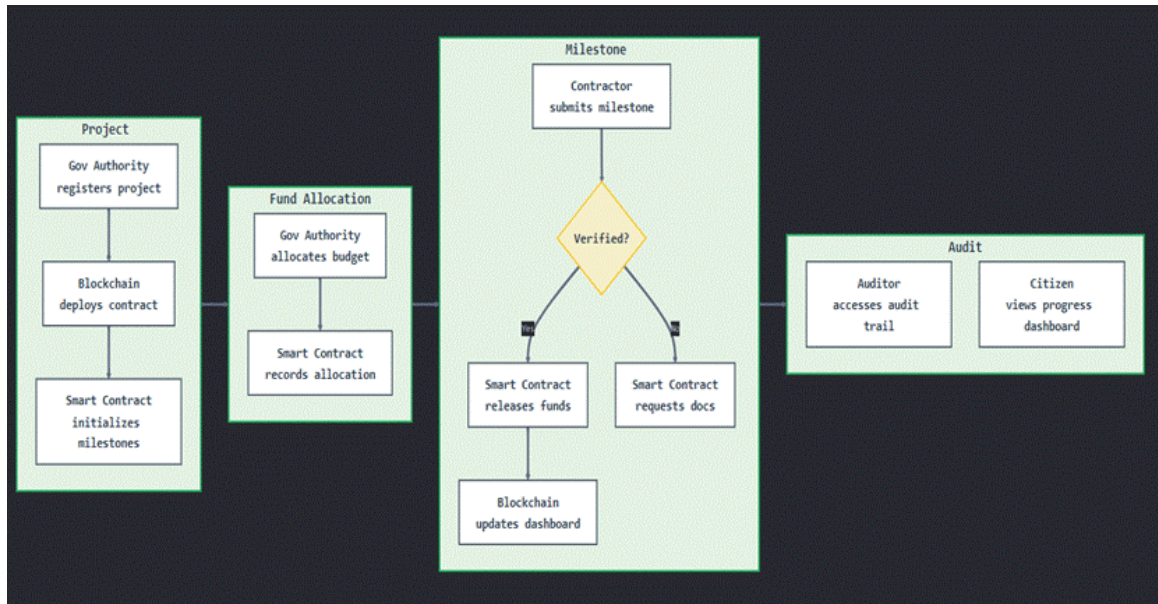


Fig 4.2.3 Sequence Diagram - system workflow

IMPLEMENTATION AND EXPERIMENTAL RESULTS

5.1 EXPERIMENTAL SETUP (OR SIMULATION)

The experimental environment was designed to simulate a realistic multi-stakeholder government project ecosystem using a permissioned blockchain framework. The setup integrates blockchain infrastructure, smart contract execution, and a web-based application layer to evaluate governance workflows, access control, and automation mechanisms.

Blockchain Framework

A permissioned blockchain network was established to manage decentralized record-keeping, identity verification, and role-based access control among participating stakeholders. The framework enables secure transaction processing, immutable data storage, and controlled participation suitable for governance-oriented applications.

Network Topology

A three-channel architecture was implemented to logically isolate data traffic and workflows related to Planning, Execution, and Financial processes. Each channel maintains a separate ledger and smart contract logic, ensuring confidentiality while enabling coordinated project governance across stakeholders.

Stakeholder Nodes

Eight distinct peer nodes were configured to represent different stakeholder entities involved in government project execution, including government ministries, contractors, auditors, financial institutions, and citizens. Each node operates under a unique organizational identity and participates in the blockchain network according to predefined access permissions and endorsement policies.

Smart Contract Environment

Smart contracts were developed to automate project lifecycle activities such as milestone validation and fund disbursement. The contracts encode predefined business rules to ensure that financial transactions are triggered only upon successful completion and verification of project milestones, thereby enforcing transparency and accountability.

Local Simulation and Testing Environment

A local blockchain simulation environment was utilized to enable rapid deployment, testing, and validation of transaction logic. This setup facilitated efficient debugging and evaluation of smart contract behavior without deploying the system on a production-scale network.

Development Tools and Application Interface

The user interface was developed using a modern web framework to provide role-specific dashboards and real-time project monitoring. The application communicates with the blockchain network through a client-side library, enabling secure interaction with smart contracts and retrieval of immutable ledger data.

5.2 EXPERIMENTAL ANALYSIS

5.2.1 DATA

The data utilized in this simulation represents a variety of government project categories, including infrastructure (roads and bridges) and social welfare initiatives.

Project Registry Data

The system processed on-chain project details such as budget allocations, project titles, and predefined timelines for eight simulated stakeholders.

Milestone & Financial Records

Data entries included cryptographic hashes of work certification reports, timestamped progress updates, and smart contract-driven fund disbursement logs.

Document Metadata

Large files like blueprints and compliance reports were stored off-chain via IPFS, with only the immutable verification hashes being processed as on-chain data.

Role-Based Access Logs

The dataset included identity management logs through Membership Service Providers (MSPs) to verify that only authorized contractors or auditors could modify specific records.

5.2.2 PERFORMANCE PARAMETERS

The system's efficiency was measured against several non-functional requirements to ensure its viability for national governance adoption.

- **Transaction Throughput:** The network was evaluated for its ability to maintain a throughput of 200–500 transactions per second (TPS) during peak simulation loads.
- **Confirmation Latency:** A critical parameter was the transaction confirmation time, which was successfully maintained at a maximum of 5 seconds to prevent bureaucratic delays.
- **System Responsiveness:** Standard user operations, such as querying project status via the dashboard, were analyzed to ensure a response time not exceeding 2–3 seconds.
- **Smart Contract Execution Accuracy:** The "condition-based" payment logic was tested to ensure 100% accuracy in releasing funds only upon the successful cryptographic verification of milestones.
- **Availability and Redundancy:** The setup was monitored to ensure a minimum uptime of 99.5%, simulating the high availability required for essential public services

5.3 WORKING OF THE PROJECT

5.3.1 PROCEDURAL WORKFLOW

Stakeholder Registration and Identity Initialization

The workflow begins with the registration of participating stakeholder organizations into the blockchain network. Each organization is issued a digital identity through the Membership Service Provider, enabling secure authentication and authorization. Users within each organization are assigned specific roles, such as planning authority, contractor, auditor, or finance officer, which determine their access rights and permissible actions within the system.

Project Creation and Planning Approval

An authorized planning authority initiates a new project by submitting project details, timelines, and budget allocations on the Planning channel. The submitted project proposal undergoes a predefined approval workflow, where relevant authorities review and approve the project parameters. All approvals are recorded immutably on the blockchain, ensuring traceability and accountability.

Task Assignment and Execution Tracking

Once the project is approved, execution-related tasks are assigned to contractors through the Execution channel. Contractors periodically submit progress updates and supporting evidence for completed tasks. These updates are recorded as blockchain transactions, enabling real-time visibility into project execution while preventing unauthorized data modification.

Inspection and Work Certification

Authorized auditors or inspectors review submitted progress reports and conduct inspections as required. Upon verification, work certification is recorded on the blockchain through smart contract execution. This certification acts as a prerequisite for financial authorization, ensuring that payments are linked directly to verified project milestones.

Financial Validation and Conditional Payment Authorization

Following successful work certification, the Financial channel smart contracts validate compliance with predefined conditions such as milestone completion, approval status, and budget limits. Once all conditions are satisfied, the system automatically authorizes payment requests. This condition-based automation reduces manual intervention and prevents unauthorized or premature fund releases.

Real-Time Monitoring and Notifications

Throughout the project lifecycle, all stakeholders can monitor project status through role-specific dashboards provided by the web-based interface. The system generates real-time notifications for approvals, milestone completions, and financial authorizations, enabling timely decision-making and coordination.

Audit and Record Preservation

All transactions, approvals, and workflow events are permanently stored on the blockchain ledger. These immutable records enable continuous auditing and post-project analysis without reliance on retrospective documentation. The audit trail ensures transparency and strengthens governance oversight.

5.3.2 ALGORITHMIC APPROACHES USED

To solve the challenges of centralization and opacity, the following technical approaches were utilized:

- **Consensus Mechanism:** A Proof of Authority (PoA) algorithm was simulated to provide high transaction throughput and low latency, which is more suitable for a permissioned government network than energy-intensive alternatives.
- **Smart Contract Logic (Chaincode):** Self-executing programs were developed to enforce "if-then" business rules, ensuring that payments are processed only when cryptographic conditions (milestone verification) are met.

- **Identity Management Algorithms:** Decentralized Identifiers (DIDs) and cryptographic signatures were used to ensure non-repudiation, meaning a stakeholder cannot deny a transaction they have authorized.
- **Hashing for Data Integrity:** SHA-256 hashing algorithms were employed to create unique fingerprints of off-chain documents, ensuring any unauthorized alteration of a project report can be detected instantly.

5.3.3 PROJECT DEPLOYMENT

The deployment environment was architected to balance security with real-world accessibility:

- **Blockchain Layer:** The core network was deployed using a local Hyperledger Fabric or Ganache simulation for the mid-semester evaluation, providing a sandbox for smart contract testing.
- **Front-End Hosting:** The user-facing dashboard, built with React.js, was deployed via Netlify to provide web access to stakeholders.
- **Gateway Integration:** Infura was utilized to provide scalable access to blockchain nodes without the overhead of maintaining a full local node.
- **Off-Chain Integration:** An IPFS node was established for document management, integrated via APIs to allow for seamless uploading and retrieval of project blueprints and media evidence.

5.3.4 SYSTEM SCREENSHOTS

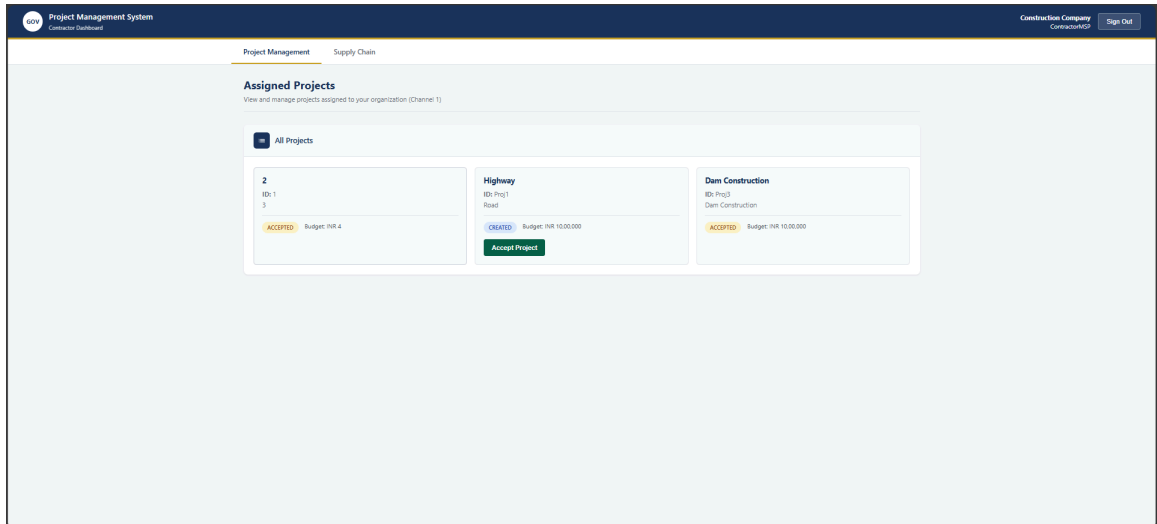


Fig 5.3.4.1 Project Management Dashboard

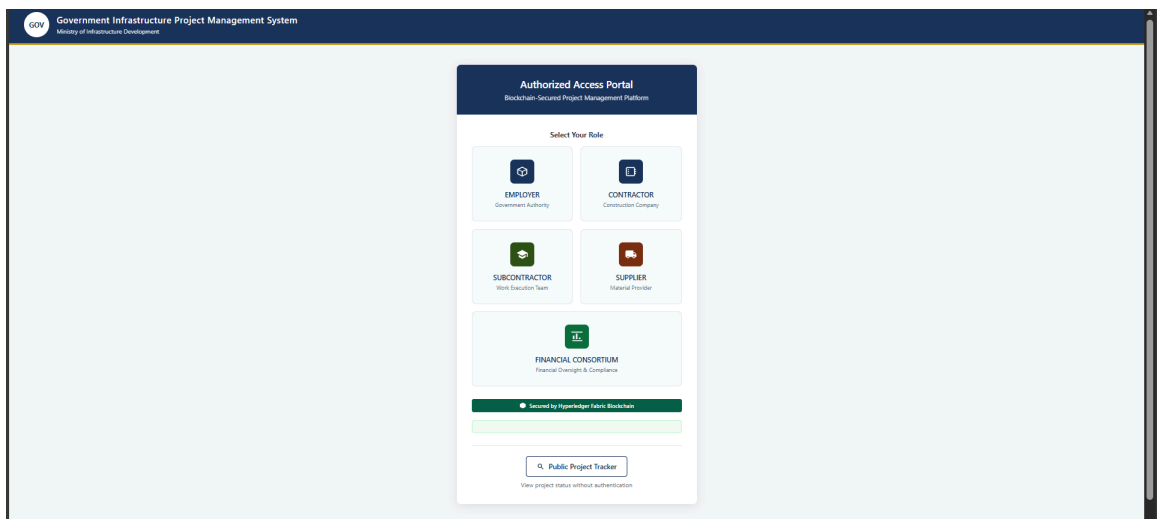


Fig 5.3.4.2 Access Portal

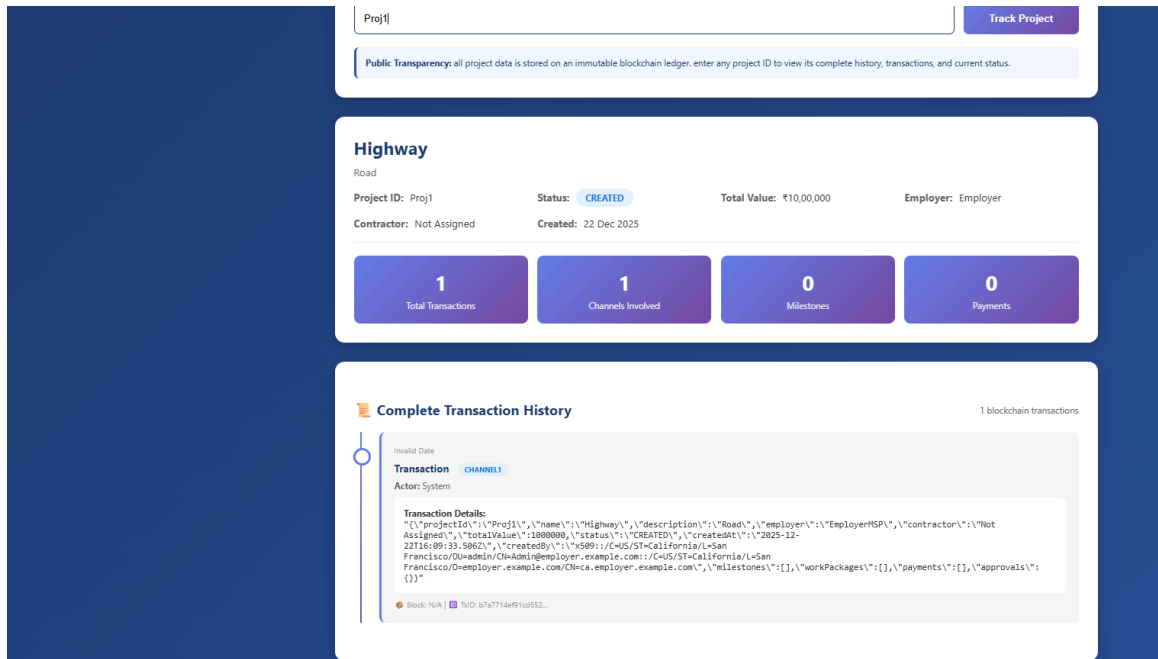


Fig 5.3.4.3 Track Project

5.4 TESTING PROCESS

5.4.1 TEST PLAN

The testing process was designed to systematically validate the functional correctness, security enforcement, and governance integrity of the proposed blockchain-based project management system. Given the distributed and permissioned nature of the system, testing focused on verifying role-based access control, smart contract execution, workflow sequencing, data integrity, and auditability rather than low-level user interface behavior.

Testing was conducted in a controlled local environment simulating a multi-stakeholder government project ecosystem. The process followed a structured approach, beginning with validation of core blockchain infrastructure components and progressing toward end-to-end workflow verification. Each testing phase was aligned with the system requirements identified during requirement analysis.

The testing process began with component-level validation, where individual smart contract functions were executed to verify correct handling of project creation, work

submission, verification, and approval logic. This phase ensured that chaincode functions behaved deterministically and enforced predefined business rules.

Subsequently, role-based testing was performed to validate identity management and access control mechanisms. Transactions were initiated using identities associated with different stakeholder roles to confirm that authorized actions were permitted and unauthorized actions were consistently rejected. This ensured correct enforcement of Membership Service Provider configurations and role-based permissions.

The testing process then progressed to workflow and integration testing, where complete project lifecycle scenarios were executed across planning, execution, and financial channels. This phase verified dependency enforcement between workflow stages, ensuring that actions such as financial approval could occur only after successful work certification. Integration between the blockchain network and the application interface was also validated to ensure consistency between user actions and ledger state.

Security and data integrity testing formed a critical part of the testing process. Certificate-based authentication, cryptographic transaction signing, and endorsement policies were validated to ensure non-repudiation and resistance to unauthorized access. Off-chain document integrity was verified through cryptographic hash comparison, confirming tamper resistance and audit trail reliability.

Finally, validation testing was conducted to assess whether the implemented system met the defined project objectives. Observations from executed test cases were analyzed to confirm transparency, accountability, workflow automation, and data immutability. Any limitations identified during testing, such as the absence of external payment gateway integration or large-scale performance evaluation, were documented as constraints rather than system failures.

Overall, the testing process confirmed that the proposed system functions correctly within its defined scope and effectively enforces governance rules in a multi-stakeholder environment.

CONCLUSIONS AND FUTURE DIRECTIONS

6.1 CONCLUSIONS

This project demonstrates the potential of blockchain technology to transform the way government projects are tracked, managed, and executed. Traditional monitoring systems suffer from inefficiencies, lack of transparency, and vulnerability to corruption, which ultimately erode public trust. By leveraging blockchain's inherent features of decentralization, immutability, and transparency, the proposed system provides a tamper-proof, automated, and inclusive framework for project monitoring.

Through the use of smart contracts, the system ensures that funds are disbursed only upon verified milestone completion, thereby reducing delays and eliminating the possibility of manual manipulation. The inclusion of role-based access control and citizen-facing dashboards fosters accountability among contractors, auditors, and officials while empowering citizens with real-time access to project progress and expenditure.

The capstone project has successfully achieved its objectives by developing a prototype that illustrates how blockchain can address key governance challenges. While the current implementation is limited in scale, it provides a strong proof of concept that can be expanded into large-scale deployments. Ultimately, the work establishes a foundation for integrating blockchain into e-governance, ensuring efficient utilization of public resources and strengthening citizen trust in government processes.

6.2 ENVIRONMENTAL, ECONOMIC AND SOCIETAL BENEFITS

Environmental

The system contributes indirectly to environmental sustainability by promoting efficient utilization of resources in government projects. With transparent fund allocation and milestone-based monitoring, wastage of materials, duplication of work, and unnecessary delays are minimized. This ensures that infrastructure and development projects are executed in a planned manner, reducing environmental degradation caused by extended

construction activities. Furthermore, by encouraging digital records and decentralized storage, the system significantly reduces reliance on paper-based documentation, thereby lowering carbon footprint.

Economic

Economically, the proposed system helps in reducing corruption, fund leakages, and mismanagement of public money. Smart contracts ensure that funds are released only when specific milestones are verified, leading to efficient use of resources and timely project completion. This minimizes cost overruns and maximizes the return on investment for government spending. In the long run, increased efficiency translates into higher savings for the exchequer, better infrastructure delivery, and improved investor confidence in public projects.

Social

From a social perspective, the project enhances trust between citizens and government authorities by providing transparent and real-time information on project progress. Citizen-facing dashboards empower communities to hold contractors and officials accountable, fostering participatory governance. This transparency builds public confidence in government institutions and reduces dissatisfaction caused by project delays or corruption. Additionally, timely completion of projects improves the quality of life by providing citizens with better infrastructure, healthcare, education facilities, and other public services.

6.3 REFLECTIONS

The development of chainZ has provided significant insights into the intersection of decentralized technology and public administration. Throughout this project, several key reflections emerged regarding the technical and operational feasibility of blockchain for e-governance:

- **Shift from Institutional to Technological Trust:** We realized that the primary value of this project is not just digitization, but the fundamental shift from relying on centralized authorities to trusting an immutable code-base. By embedding

transparency into the technological core, we effectively addressed the "Principal-Agent" problem where agents might otherwise act in self-interest.

- **The Power of Automation via Smart Contracts:** Implementing smart contracts for fund disbursement highlighted how much administrative latency is caused by manual verification. Transitioning to cryptographic verification of milestones proved that bureaucratic bottlenecks can be significantly minimized while reducing the risk of financial misappropriation.
- **Balancing Transparency with Data Privacy:** A critical reflection during the design phase was the tension between public transparency and the need for data privacy. Adopting a hybrid blockchain model—using private channels for sensitive financial data while keeping general project progress public—was a vital design choice to ensure compliance with data protection standards like GDPR and the Personal Data Protection Bill.
- **Integration Challenges with Legacy Infrastructure:** We observed that while blockchain offers a superior alternative, its immediate large-scale adoption is constrained by the need to interoperate with existing, often fragmented, government IT systems. This underscores the importance of our modular architecture and the use of standardized APIs for future-proofing the system.
- **Empowering Participatory Governance:** Finally, the development of citizen-facing dashboards reinforced the idea that transparency is a prerequisite for public trust. Providing citizens with real-time, verifiable access to how public funds are utilized transforms them from passive beneficiaries into active stakeholders in the governance process.

6.4 FUTURE WORK

1. **User Testing and Refinement:** Conduct User Acceptance Testing (UAT) to validate dashboard and document management features. Refine UI/UX based on stakeholder feedback and improve usability.
2. **Scalability and Performance Evaluation:** Perform load testing to ensure the system can handle multiple concurrent projects and users. Optimize system performance for large-scale deployments.

3. **IoT Integration for Automated Verification:** Incorporate Internet of Things (IoT) devices, such as sensors, to provide automated, real-time data collection from construction sites. This would allow smart contracts to trigger milestone approvals based on physical data (e.g., structural completion) rather than relying solely on manual reports.
4. **Cross-Chain Interoperability:** Explore frameworks to enable interoperability between the current permissioned network and other blockchain ecosystems. This would facilitate secure data exchange with international financial institutions or multi-national regulatory bodies.
5. **AI-Driven Predictive Analytics:** Policy makers could use the feature of machine learning modules to analyse the previous projects data stored on the blockchain to identify possible cost overruns, potential bottlenecks before they occur and make changes accordingly.

7.1 CHALLENGES FACED

A major challenge was the lack of thorough documentation and open-source project samples for Hyperledger Fabric. The scarcity of existing, large-scale implementation examples made it intricate to find practical examples to use as a baseline for the system's development.

Another challenge was building a decentralized system that could potentially combine with existing, paper-based records and fragmented government databases.

To ensure that various types of projects could be handled by the fund disbursement logic of the system, rigorous testing and formal verification was required.

To maintain the boundary between the data that could be made public and the one which was sensitive like financial details, a hybrid model was integrated which used complex channel configurations within blockchain.

Operational risk of wrong entry of data like falsified information or inaccurate data was encountered as once the data is entered into blockchain, it's immutable.

7.2 RELEVANT SUBJECTS

- Blockchain and distributed ledger technologies are used to build a decentralized system, consensus mechanisms, and create records that can't be changed.
- Cryptography and network security help with managing user identities, safely sending data through the internet and digital signatures for non-repudiation.
- Software Engineering and SRS are crucial for outlining the functional requirements, designing its structure, and modular design patterns.
- Database management systems are needed for the off-chain storage of large project datasets and to connect them with the on-chain ledger.

- Web development, both front-end and back-end is used to build the React.js frontend and set up tools that let the users and stakeholders interact with the blockchain.

7.3 INTERDISCIPLINARY KNOWLEDGE SHARING

This project bridged multiple domains, requiring knowledge exchange between technical and non-technical fields:

- **Public Administration and Governance:** Research about current government project monitoring workflows and the ‘Principal-Agent’ theory helped find the problems with transparency in the system.
- **Legal and Regulatory Frameworks:** Studying data protection laws like GDPR and the Personal Data Protection Bill helped build the system’s privacy protection features.
- **Economics and Finance:** Understanding financial auditing and how money is released was important to program the smart contracts that could send payments when conditions are met.
- **Environmental Science:** Using knowledge about managing resources and carbon footprints, the evaluation of the environmental benefits of reducing paper-based documentation was done.

7.4 PEER ASSESSMENT MATRIX

		Evaluation of				
		Aditya	Navyaa	Garima	Amolik	Shankhadeep
Evaluation By	Aditya	-	4	3	5	5
	Navyaa	4	-	3	5	5
	Garima	4	4	-	5	5
	Amolik	4	3	4	-	4

	Shankhadeep	3	4	5	4	-
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Table 7.4.1 : Peer assessment matrix

7.5 ROLE PLAYING AND WORK SCHEDULE

Roll Number	Team Member	Role
102203083	Aditya Singh	Front end and back end
102203276	Navyaa Gupta	Network Architecture and Configuration
102203296	Garima Singh	Blockchain integration
102203858	Amolik Janjua	Channel creation with chaincode
102203883	Shankhadeep Maiti	Role based endorsement

Table 7.5.1 : Role playing and work schedule

7.6 STUDENT OUTCOMES DESCRIPTION AND PERFORMANCE INDICATORS

SO	SO Description	Outcome
1.1	Able to identify and formulate problems related to computational domain	Recognized the issue of lack of transparency and too much control in tracking public projects. Created a system that spreads out control to prevent misuse of funds and missed deadlines.
1.2	Apply engineering, science and mathematics body of knowledge to obtain analytical, numerical, and statistical solutions to solve engineering problems	
2.1	Design computing system(s) to address needs in different problem domains and build prototypes, simulations, proof of concepts, wherever necessary, that meet	Created a system with several layers that includes a Hyperledger Fabric network and dashboards based on user roles. Developed a

	design and implementation specifications.	working model that can automatically check project milestones and release funds.
2.2	Ability to analyze the economic trade-offs in computing systems.	
3.1	Prepare and present variety of documents such as project or laboratory reports according to computing standards and protocols.	Wrote a detailed report that explains the system's structure, data flow diagrams, and sequence diagrams. Made sure all the documents met standard academic and technical guidelines.
3.3	Able to communicate effectively with peers in well organized and logical manner using adequate technical knowledge to solve computational domain problems and issues.	
4.1	Aware of ethical and professional responsibilities while designing and implementing computing solutions and innovations.	Added rules to stop unfair practices and make sure money is used as planned. Kept important information safe by using secret channels that protect personal details.
4.3	Evaluate computational engineering solutions considering environmental, societal, and economic contexts.	
5.1	Participate in the development and selection of ideas to meet established objective and goals.	
5.2	Able to plan, share and execute task responsibilities to function effectively by creating collaborative and inclusive environment in a team.	Collaborated effectively in a team to execute a Work Breakdown Structure, dividing tasks between backend blockchain development and frontend UI implementation.
6.1	Ability to perform experimentations and further analyze the obtained results.	
6.2	Ability to analyze and interpret data, make necessary judgement(s) and draw conclusion(s).	Performed pilot simulations to evaluate transaction throughput and latency. Analyzed results to

		confirm that automated smart contracts triggered correctly upon milestone completion.
7.1	Able to explore and utilize resources to enhance self-learning.	Independently researched papers and official documentation to overcome the lack of Hyperledger project examples. Successfully learned to develop and deploy smart contracts through self-directed study..

Table 7.6.1 Student outcomes description

7.7 BRIEF ANALYTICAL ASSESSMENT

The capstone project successfully met its intended objectives by designing and implementing a blockchain-based governance framework for monitoring government projects. The system effectively demonstrates how decentralization, role-based access control, and smart contract automation can address key challenges such as lack of transparency, delayed approvals, and misuse of public funds.

Despite being a prototype-level implementation, the project validates the feasibility of applying permissioned blockchain technology in e-governance scenarios. The integration of automated workflows, immutable audit trails, and real-time dashboards highlights the system's potential to improve accountability, efficiency, and stakeholder trust.

Overall, the project reflects strong technical understanding, effective teamwork, and practical problem-solving skills, making it a successful and relevant academic contribution with clear scope for real-world extension.

APPENDIX A : REFERENCES

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APPENDIX B : PLAGIARISM REPORT

cp919 HSpenny

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20	Aamir, Shaheen. "Consortium Blockchain in E-Government & Smart Democracy: Learning from the Past for the Future: How the Data More Effective and Secure in the Context of Consortium Blockchain", Universidade NOVA de Lisboa (Portugal), 2024 Publication	<1 %
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26	Ghulam Mustafa, Waqas Rafiq, Naveed Jhamat, Zeeshan Arshad, Farhana Aziz Rana. "Blockchain-based governance models in e- government: a comprehensive framework for legal, technical, ethical and security considerations", International Journal of Law and Management, 2024 Publication	<1 %
27	Gideon Adjorlolo, Zhiwei Tang, Gladys Wauk, Philip Adu Sarfo, Alhassan Baako Braimah, Richard Blankson Safo, Benedict N-yanyi. "Evaluating Corruption-Prone Public Procurement Stages for Blockchain Integration Using AHP Approach", Systems, 2025 Publication	<1 %

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